

Software Engineering (UE24CS341A)

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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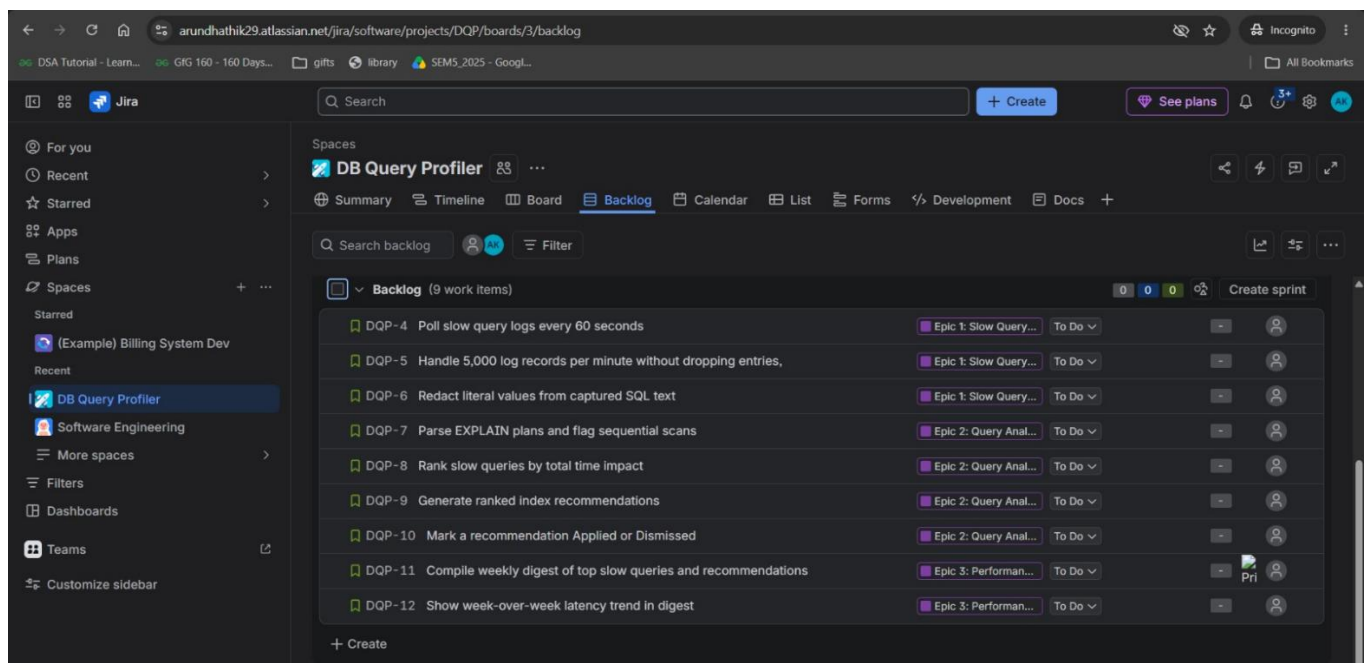
Project: Database Query Performance Profiler (Problem Statement #44)

Jira Project Link:

<https://arundhathik29.atlassian.net/jira/software/projects/DQP/boards/3/backlog>

This document contains the Jira evidence for Lab 2: the Epics and User Story backlog, story point estimates, an active sprint board, the sprint burndown chart, and reflection answers, covering two completed sprints as instructed.

1. Epics & User Stories / Backlog



Backlog view: 3 Epics (Slow Query Ingestion & Monitoring, Query Analysis & Index Recommendations, Performance Reporting & Digest) with 9 User Stories mapped from the Lab 1 functional requirements, each tagged to its parent Epic.

2. Story Points

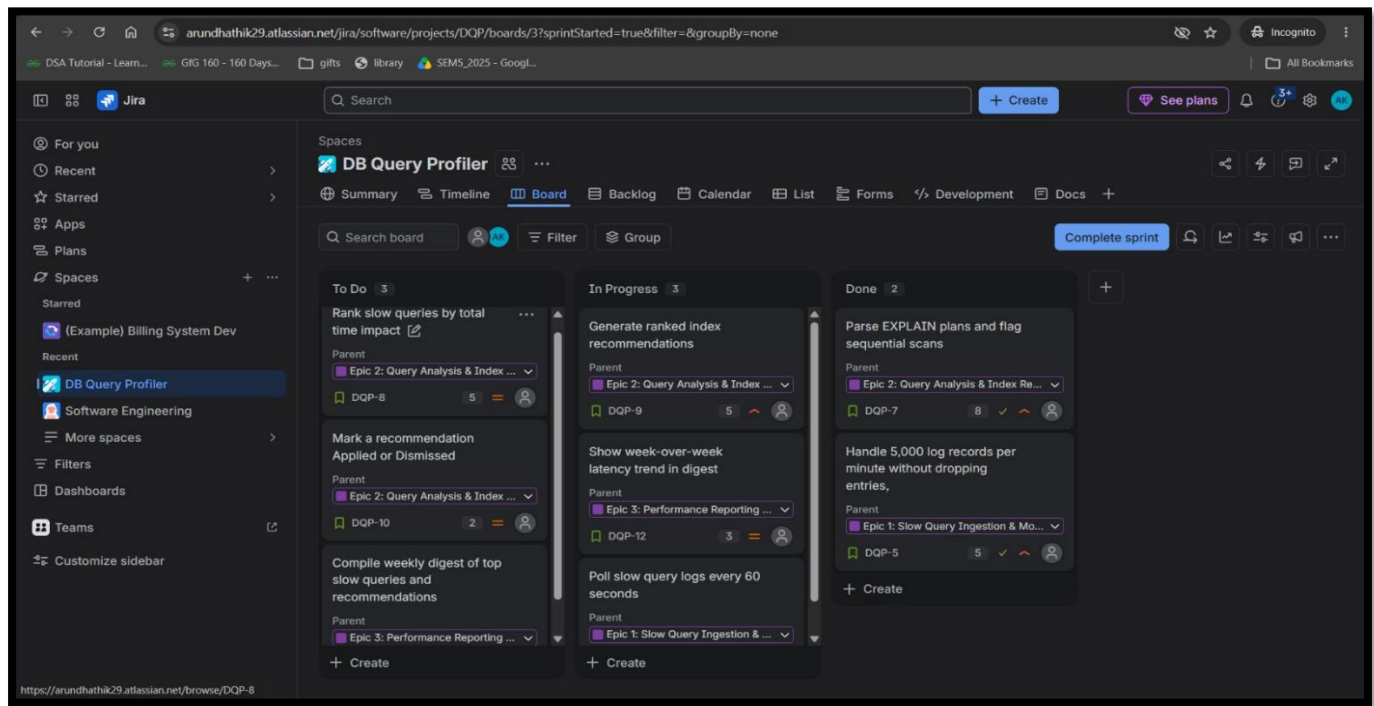
The screenshot shows the Jira interface for the 'DB Query Profiler' space. The 'Backlog' tab is selected, displaying a list of 12 tasks. Each task is associated with an epic and has a story point estimate. The total backlog estimate is 38 points.

Task ID	Task Description	Epic	Priority	Story Points
DQP-4	Poll slow query logs every 60 seconds	Epic 1: Slow Query...	To Do	2
DQP-5	Handle 5,000 log records per minute without dropping entries,	Epic 1: Slow Query...	To Do	5
DQP-6	Redact literal values from captured SQL text	Epic 1: Slow Query...	To Do	3
DQP-7	Parse EXPLAIN plans and flag sequential scans	Epic 2: Query Anal...	To Do	8
DQP-8	Rank slow queries by total time impact	Epic 2: Query Anal...	To Do	5
DQP-9	Generate ranked index recommendations	Epic 2: Query Anal...	To Do	5
DQP-10	Mark a recommendation Applied or Dismissed	Epic 2: Query Anal...	To Do	2
DQP-11	Compile weekly digest of top slow queries and recommendations	Epic 3: Performan...	To Do	5
DQP-12	Show week-over-week latency trend in digest	Epic 3: Performan...	To Do	3

All 9

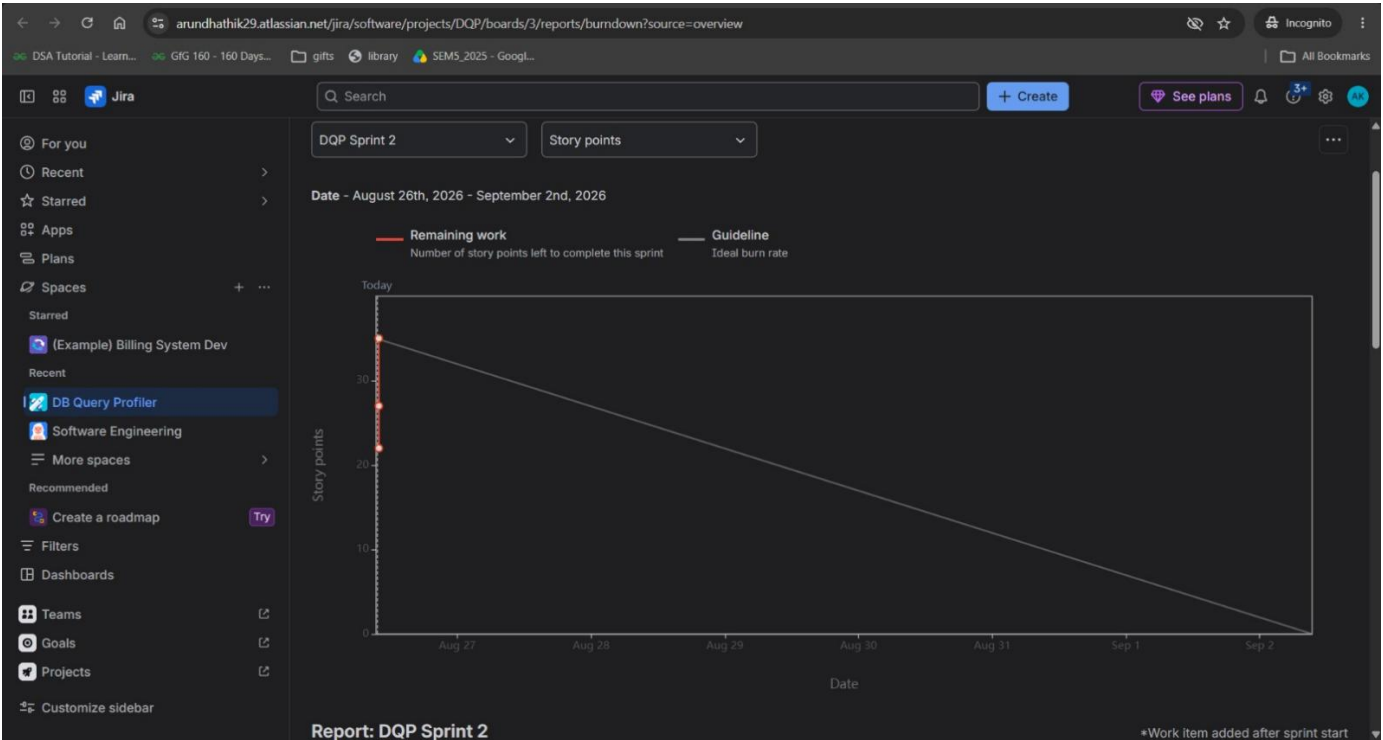
stories with Priority (High/Medium) and Story Point estimates (Fibonacci scale) assigned. Total backlog estimate: 38 points.

3. Active Sprint Board



Sprint 2 in progress: 3 items In Progress, 2 items Done, 3 items still To Do, spanning stories from all three Epics.

4. Burndown Chart



Sprint burndown chart for DQP Sprint 2, showing story points remaining against the ideal burn-rate guideline.

5. Reflection Questions

Fill in each answer below based on what you actually observed while running Sprint 1 and Sprint 2

1. Did your estimations reflect the actual effort?

Mostly yes, but not perfectly. Simple stories like redacting literal values or marking a recommendation Applied/Dismissed were correctly estimated low (2-3 points), while more complex work like parsing EXPLAIN plans got a higher estimate (8 points), which felt appropriate given how many sub-steps it involves. However, Sprint 1 showed a gap: two of the three planned stories (DQP-4, DQP-5) weren't finished within the sprint and had to carry over to Sprint 2, suggesting the ingestion-related stories were slightly under-estimated for the time available.

2. Was your backlog well-prioritized?

Largely yes. The High-priority stories mapped directly to the core FRs (ingestion, EXPLAIN plan parsing, index recommendations, weekly digest), which matched their importance to the overall system. Medium-priority stories (ranking by impact, marking recommendations, latency trend) were genuinely lower-stakes, supporting features rather than core functionality. In hindsight, "Redact literal values" (a security requirement) probably deserved to be prioritized earlier, since it protects sensitive data from day one.

3. How did your simulated sprint align with your plan?

The plan was to finish all of Epic 1 in Sprint 1, but only one of three stories reached Done; the rest were still In Progress or To Do when the sprint was completed. That's realistic, it's common for a team's first sprint to overcommit. Sprint 2 absorbed the leftover work from Sprint 1 alongside the Epic 2 and Epic 3 stories, which mirrors how real teams re-plan around carried-over work rather than treating the original plan as fixed.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart's remaining-work line started well above the ideal guideline line and needed to drop sharply to finish on schedule, which visually confirmed that the sprint was overloaded relative to the time box. This suggests that for future sprints, either the team

should commit to fewer story points per sprint, or break down the larger stories (like the 8-point EXPLAIN plan parsing) into smaller, more predictable chunks.